Which of the following pairs of enzymes are involved in the release of individual monomers from glycogen?

I. Phosphoglucomutase

II. Glucose 6-phosphatase

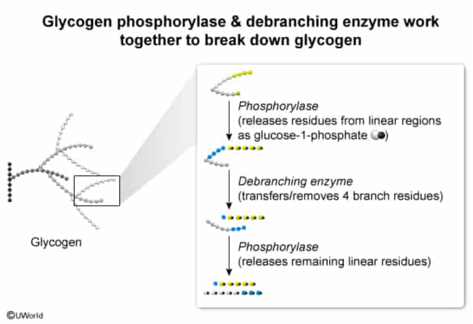
III. Glycogen phosphorylase

IV. Debranching enzyme

- A. I and II only [5%]
- B. I and III only [8%]
- C. II and IV only [18%]
- ✔ D. III and IV only [68%]

Correct
68% answered correctly

Explanation:



Glycogenolysis provides glucose molecules for use in other pathways (eg, glycolysis, the pentose phosphate pathway) in the tissue in which glycogenolysis occurs. In addition, the liver possesses enzymes allowing the glucose 1-phosphate formed by glycogenolysis to be converted in subsequent reactions into unmodified glucose. This liver-derived glucose can then be released into the bloodstream for use by other tissues.

The question asks for the pair of enzymes involved in the release of monomers from glycogen. **Phosphorylase** acts upon **linear glycogen** regions, **catalyzing** the **release** of **glucose residues** from the **nonreducing end** (ie, the end with a glucose molecule that only participates in one glycosidic bond) **(Number III)**. **Debranching enzyme "linearizes" branched regions** of glycogen, repositioning them onto the ends of short linear segments and releasing one glucose residue from the branch in the process **(Number IV)**.

(Number I) Phosphoglucomutase converts the glucose 1-phosphate released by glycogenolysis into glucose 6-phosphate. Therefore, phosphoglucomutase acts on a *product* of monomer production rather than *producing* those monomers.

(Number II) Glucose 6-phosphatase converts glucose 6-phosphate to glucose, allowing glucose stored in liver glycogen to be released into circulation. Because this process occurs downstream of glycogenolysis, glucose 6-phosphatase is not involved in the production of monomers from glycogen.

Educational objective:
Glycogenolysis releases glucose residues from glycogen. Glycogen phosphorylase catalyzes the release of glucose residues from linear glycogen segments, and debranching enzyme linearizes branched segments of glycogen for subsequent degradation by phosphorylase.

Time spent: 0 sec	**QID:** 403530
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